# Reservoir Sampling

## Uniform Sampling from a Stream

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# Problem

Items $x_1, x_2, \ldots, x_m$ arrive one by one. $m$ is unknown.

**Goal**: maintain a uniform random sample of size $k$ using $O(k)$ memory.

# Algorithm R (Vitter 1985)

1. Store the first $k$ items in reservoir $R$.
2. For item $x_i$ with $i > k$:
   2.1 Draw $j \sim \mathrm{Uniform}\{1, \ldots, i\}$.
   2.2 If $j \leq k$, set $R[j] \leftarrow x_i$.

# Correctness

**Claim**: after $i$ items, each is in $R$ with probability $k/i$.

**Inductive step**: item $x_{i+1}$ arrives.

- $x_{i+1}$ enters w.p. $k/(i+1)$.
- Existing $x_s$ survives (by induction it is in $R$ w.p. $k/i$):

$$\Pr(x_s \text{ evicted}) = \underbrace{\frac{k}{i+1}}_{x_{i+1} \text{ enters}} \cdot \underbrace{\frac{1}{k}}_{\text{hits slot of } x_s} = \frac{1}{i+1}.$$

$$\Pr(x_s \text{ survives}) = \frac{k}{i} \cdot \left(1 - \frac{1}{i+1}\right) = \frac{k}{i} \cdot \frac{i}{i+1} = \frac{k}{i+1}.$$

# Complexity

- **Space**: $O(k)$
- **Time**: $O(1)$ per item
- **Passes**: 1

**Algorithm Z**: skip items geometrically; reduces draws to $O(n/k)$ total.

# Second Frequency Moment $F_2$

$$F_2 := \sum_i f_i^2.$$

- 1,1,1,1,1,1. $F_2 = 36$.
- 1,1,2,2,3,3. $F_2 = 12$.
- 1,2,3,4,5,6. $F_2 = 6$.

$F_2$ measures how skewed the stream is.

# Frequency Moments

$$F_k := \sum_i f_i^k.$$

# AMS Sampling Idea

Goal: estimate frequency moments in one pass.

Sample a random $x_t$ using reservoir sampling.

$$R = \left|\{t \geq r : x_t = a\}\right|.$$

For example,

$$\text{stream} = 1, 4, 3, 2, \underbrace{1}_{x_t}, 2, 3, 1, 2, 4, 3, 1, 4.$$

$$R = 3.$$

# AMS Sampling

Estimator:

$$X = m(R^k - (R-1)^k).$$

Then $\mathbb{E}[X] = F_k$.

# Why $\mathbb{E}[X] = F_k$ (1/2)

Let item $j$ appear $f_j$ times, at positions $p_{j,1} < \cdots < p_{j,f_j}$.
Since $r$ is uniform in $\{1, \ldots, m\}$, the $\frac{1}{m}$ and the $m$ in $X$ cancel:

$$\mathbb{E}[X] = \frac{1}{m} \sum_{r=1}^{m} m\big(R(r)^k - (R(r)-1)^k\big) = \sum_{r=1}^{m} \big(R(r)^k - (R(r)-1)^k\big).$$

If $r = p_{j,t}$, then $R = f_j - t + 1$. Grouping by item $j$ (item $j$ contributes exactly $f_j$ of the $m$ terms):

$$\mathbb{E}[X] = \sum_j \sum_{t=1}^{f_j} \Big((f_j - t + 1)^k - (f_j - t)^k\Big).$$

# Why $\mathbb{E}[X] = F_k$ (2/2)

The inner sum telescopes:

$$\begin{aligned}
&t = 1: \quad f_j^k - (f_j - 1)^k \\
&t = 2: \quad (f_j - 1)^k - (f_j - 2)^k \\
&\vdots \\
&t = f_j: \quad 1^k - 0^k
\end{aligned}$$

All intermediate terms cancel, leaving $f_j^k$:

$$\mathbb{E}[X] = \sum_j f_j^k = F_k. \qquad \square$$

# AMS Sampling

- Single sample is unbiased but high variance.
- Average multiple independent copies to reduce variance.
- To sample multiple items, use reservoir sampling.